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Research paper

A meta-analysis of the intervention effect of cognitive behavioral therapy on adult ADHD

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ABSTRACT

Objective: This meta-analysis systematically evaluated the efficacy of Cognitive Behavioral Therapy (CBT) for adult Attention-Deficit/Hyperactivity Disorder (ADHD).
Methods: A systematic search was conducted across PubMed, Medline, and the Cochrane Library to identify relevant randomized controlled trials (RCTs). Methodological quality assessment and statistical analysis were performed using Review Manager 5.3 in accordance with Cochrane guidelines.
Results: Analysis of 14 RCTs selected from 949 identified records showed significant benefits of CBT for core ADHD symptoms (SMD = −0.45, $P < 0.00001$), depressive mood (SMD = −0.23, $P = 0.001$), anxiety/stress (SMD = −0.24, $P = 0.003$), and executive function (SMD = −0.43, P = 0.001). No significant effects were observed for quality of life (SMD = 0.12, $P = 0.47$) or self-evaluation (SMD = −0.30, $P = 0.22$). Prespecified subgroup analyses revealed distinct efficacy profiles. By treatment composition, pure CBT demonstrated superior effects on core symptoms (SMD = −0.51, $P < 0.00001$) and executive function (SMD = −0.56, $P < 0.0001$), while CBT combined with medication showed greater efficacy for quality of life (SMD = 0.34, $P = 0.02$) and anxiety/stress (SMD = −0.53, $P = 0.0004$). By delivery mode, efficacy varied symptom-specifically: group-based formats (Group CBT, Specialized Skills Training) showed stronger effects on core symptoms, whereas Individual CBT was most effective for improving emotional outcomes (depressive mood, anxiety/stress) and quality of life.
Conclusion: CBT demonstrates efficacy for core and associated symptoms of adult ADHD, with the optimal intervention strategy varying meaningfully by outcome and modality. A symptom-guided, stepped-care approach is supported: group-based CBT formats are preferable for targeting core symptoms and executive function, while individual CBT better addresses emotional dysregulation and quality of life. The combination of CBT with medication offers added benefit for anxiety and quality of life. The consistent lack of effect on self-evaluation highlights a critical unmet therapeutic need.

1. Introduction

Attention-deficit/hyperactivity disorder (ADHD), one of the most common neurodevelopmental disorders of childhood (Faraone et al., 2024; Safren et al., 2010), persists into adulthood in approximately 30–50 % of cases (Wang, 2019; American Psychiatric Association, 2013). While hyperactive symptoms often diminish with age, core deficits of inattention and impulsivity frequently continue (Majarwitz and Perumareddi, 2023; Nakashima et al., 2022; Wu et al., 2021). These are compounded in adulthood by significant challenges in executive functions, including planning, organization, and time management, alongside symptoms of restlessness, emotional lability, and impulsive behavior (Faheem et al., 2022; Anbarasan et al., 2020; Wymbs et al., 2021; Sibley et al., 2023). Furthermore, adults with ADHD commonly experience emotional dysregulation, mood instability, irritability, and low self-esteem (Schiweck et al., 2021; Anbarasan et al., 2020; Wymbs et al., 2021; Adamou et al., 2021).

The persistence of this multifaceted symptomatology profoundly impacts daily functioning, making comprehensive improvement through any single intervention challenging. Although pharmacotherapy remains a first-line treatment, growing evidence underscores the value of non-pharmacological approaches, particularly cognitive-

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behavioral therapy (CBT), for ameliorating symptoms and enhancing functioning in adults with ADHD (Sibley et al., 2023; Lopez et al., 2018).

CBT is posited to confer benefits by modifying maladaptive thought patterns and behaviors. In adult ADHD, it targets specific deficits such as poor organizational skills, time management difficulties, and emotional dysregulation (Stallard, 2022). Models like Safren's modular CBT aim to improve executive functions and emotional regulation through a combination of behavioral skill training and cognitive restructuring (Wu et al., 2021). Evidence suggests that CBT not only reduces core ADHD symptoms but also alleviates co-occurring depression and anxiety, potentially by altering patients' self-perceptions and attitudes, thereby reducing emotional distress (Liu et al., 2023). Intriguingly, neuroimaging studies indicate that CBT may share common neural mechanisms with pharmacotherapy, such as modulating connectivity in the frontoparietal network and cerebellum, to improve core symptoms (Schönenberg et al., 2017).

In clinical practice, CBT is delivered in various formats—including individual, group, internet-based, and dialectical behavior therapy (DBT). However, the comparative efficacy of these different modalities remains unclear. Moreover, the synergistic effects of combining CBT with other treatments, such as pharmacotherapy or metacognitive therapy, have not been comprehensively evaluated. Although the overall efficacy of CBT is established, key questions regarding its optimal implementation persist. A crucial yet under-explored issue is the mechanism of benefit when CBT is combined with medication: does the rapid improvement in attention from medication create a superior "learning state" for CBT skill acquisition, or do the overlapping mechanisms lead to resource competition, potentially undermining the long-term benefits of CBT? Existing meta-analyses have predominantly focused on establishing the overall effect of CBT, overlooking this critical question of treatment efficiency and personalization.

To address this, we proposed the central hypothesis that while pure CBT would be superior for targeted improvement of executive function, combination therapy would yield a synergistic effect on quality of life and anxiety, facilitated by the stabilizing effect of medication on the emotional foundation. We tested this by conducting pre-specified subgroup analyses to systematically compare the effect patterns of these two intervention pathways across different symptom dimensions, aiming to provide precise evidence for clinical decision-making.

2. Materials and methods

2.1. Study design

This systematic review and meta-analysis was conducted to evaluate and compare the efficacy of various forms of cognitive-behavioral therapy (CBT), both as a standalone intervention and in combination with other treatments, for adults with ADHD. The study protocol was registered on the PROSPERO platform (Registration ID: CRD42024583562) and the review was performed in accordance with the Cochrane Handbook for Systematic Reviews of Interventions and reported following the PRISMA guidelines.

2.2. Search strategy

A systematic literature search was performed in the following electronic databases from their inception until 2025: PubMed, Medline, the Cochrane Central Register of Controlled Trials (CENTRAL), and PsycINFO. The search strategy was designed to identify all relevant randomized controlled trials (RCTs) concerning Cognitive Behavioral Therapy for adult ADHD. It combined keywords and controlled vocabulary terms (e.g., MeSH in PubMed) related to three core concepts: "Adult ADHD," "Cognitive Behavioral Therapy," and "Randomized Controlled Trial." The specific search strategy for PubMed is presented as an example: (("Adult Attention Deficit Hyperactivity Disorder" [Mesh] OR "Attention Deficit Disorder with Hyperactivity" [Mesh] OR "adult adhd" [tiab] OR "adult attention deficit hyperactivity disorder" [tiab]) AND ("Cognitive Behavioral Therapy" [Mesh] OR "Psychotherapy, Cognitive" [Mesh] OR "cognitive behavioral therapy" [tiab] OR "CBT" [tiab]) AND ("Randomized Controlled Trial" [ptyp] OR "randomized controlled trial" [tiab] OR "RCT" [tiab])).

2.3. Inclusion and exclusion criteria

The inclusion and exclusion criteria for literature selection will be based on four aspects: study type, study population, interventions, and outcome indicators, as detailed in Table 1.

2.4. Literature screening and data extraction

The literature screening and data extraction process was conducted independently by two reviewers. Any discrepancies were resolved through discussion or by consulting a third reviewer. Initially, titles and abstracts were screened to exclude irrelevant studies. The full texts of the remaining articles were then assessed against the inclusion and exclusion criteria.

A standardized data extraction form was used to collect the following information from each included study: first author, publication year, country, sample size, participant demographics (e.g., age, gender), detailed characteristics of the intervention and control conditions (including type, frequency, and duration), and all relevant outcome data. The extracted data are summarized in Table 2 and Table 3.

2.5. Quality assessment

The methodological quality and risk of bias of the included RCTs were assessed independently by two reviewers using the Cochrane Risk of Bias tool (RoB 2.0) within the Review Manager 5.4 software. The assessment covered key domains, including selection bias (random sequence generation and allocation concealment), performance bias, detection bias, attrition bias, reporting bias, and other potential sources of bias. Each study was judged to have a "low," "high," or "unclear" risk of bias for each domain.

2.6. Statistical methods

The statistical synthesis for meta-analysis was conducted using Review Manager (RevMan, version 5.4.1, The Cochrane Collaboration).

Table 1
Inclusion and exclusion criteria for literature review.

Item	Inclusion criteria	Exclusion criteria
Study Type	Randomized controlled trials (RCTs)	Non-randomized controlled trials: such as observational studies, case reports, systematic reviews, etc.
Study Subjects	Adults with ADHD aged 18 and above	Non-adult ADHD patients: studies involving children, adolescents, or adults outside a specific age group.
Interventions	Studies including at least one form of psychological therapy intervention as the primary intervention, combined with pharmacotherapy.	Primary intervention is not a psychological therapy intervention or does not include pharmacotherapy.
Outcome Measures	Studies providing data for at least one of the following outcome measures: symptom improvement scale scores (e.g., ADHD rating scales), functional improvement scale scores (e.g., social and occupational functioning), psychological therapy-related scale scores.	Unclear outcome measures, insufficient outcome data provided, or inability to extract relevant outcome measures from the study.

Table 2
Basic information of included studies.

Included study	Country	Sample size	Gender ratio	Age	Intervention measures		Intervention duration
					Experimental	Control	
Van 2019	Netherlands	119	Male 83.2 % Female 16.8 %	18–65	CBT + Comprehensive Treatment	CBT + Substance Use Disorder Treatment	15 times in the experimental, 10 times in the control
Nasri 2023	Sweden	104	Female 59.6 %, Male 40.4 %	18–65	iCBT	iART	12 weeks
Solanto 2008	America	88	Male 29 % Female 71 %	18–65	MCT	SG	12 weeks
Moritz 2021	Brazil	31	\	18–60	DBT-ST	TaU	12 weeks
Huang 2019	China	108	Male 60.8 % Female 39.2 %	18–39	CBT/CBT + Enhancement	Waitlist Group	CBT12 weeks, Intensive treatment from weeks 13 to 24
Cherkasova 2020	Canada	88	Male 70.1 % Female 29.9 %	18–66	CBT + M	CBT	12 weeks
Lam 2019	Germany	131	Male 49.8 % Female 50.2 %	±36.3	GPT + MED	CM + MPH	1 year
Halmøy 2022	Norse	121	Male 44 % Female 56 %	±37	DBT-bGT	TAU	14 weeks
Pan 2022	China	98	Male 55 %, Female 45 %	18–45	CBT + MED	MED	12 weeks
Nakashima 2022	Japan	48	Male 12.5 % Female 88.0 %	20–65	gCBT	TAU	8 weeks
Young 2015	Britain	95	Male 35 % Female 65 %	18–73	CBT + MED	TAU+MED	8 weeks
Virta 2010	Finland	39	Male 48 % Female 52 %	18–49	CBT/CT	No intervention	10 weeks
Safren 2010	America	43	M-55 % F-45 %	18<	CBT	RWS	12 weeks
Corbisiero 2018	Switzerland	43	M-55 % F-45 %	18–49	CBT + MED	SCM + MED	12 weeks

Note: CBT-Cognitive Behavioral Therapy; TAU-Treatment as Usual; MED-Medication; MCT-Meta-Cognitive Therapy; SG-Support group; CBT + MED-Cognitive Behavioral Therapy plus Medication; SCM + MED-Standard Clinical Management plus Medication; GPT-Cognitive Behavioral Group Psychotherapy; DBT-ST-Dialectical Behavior Therapy Skills Training; gCBT-Group Cognitive Behavior Therapy; CM-Clinical Management; DBT-bGT-Dialectical Behavior Therapy Basic Group Therapy; CT-Cognitive Training; RWS-Relaxation With Educational Support; M-Male; F-Female.

The effect size was calculated as the Standardized Mean Difference (SMD) alongside its 95 % confidence interval (CI). The degree of heterogeneity among the included studies was quantified using the I^2 statistic and formally tested with Cochran's Q test. Based on these metrics, the data were pooled using a fixed-effect model under conditions of low heterogeneity ($I^2 \leq 50$ % and *P*-value for Q test ≥ 0.1) or a random-effects model in the presence of significant heterogeneity ($I^2 > 50$ % and $P < 0.1$).

For enhanced visual representation, a hybrid analytical workflow was adopted. The numerical results derived from RevMan, encompassing study-specific point estimates, confidence intervals, and weights, served as the input for specialized visualization routines. All forest plots and funnel plots were programmatically created using MATLAB (R2024a, The Math Works, Inc.). This strategy leveraged the robust statistical framework of RevMan for data synthesis while utilizing the advanced graphical capabilities of MATLAB, ensuring the production of high-fidelity, customizable figures that meet stringent publication standards.

2.7. Subgroup and sensitivity analyses

To explore potential sources of heterogeneity and examine the potential influence of intervention characteristics, pre-specified subgroup analyses were performed. In addition to the primary comparison of pure CBT versus CBT combined with pharmacotherapy, we conducted an exploratory subgroup analysis based on the mode of CBT delivery. Studies were categorized into four mutually exclusive groups: 1. Individual CBT: Standard or modular CBT delivered in a one-on-one, therapist-patient format. 2. Group CBT: CBT delivered in a group setting, including group-based metacognitive therapy. 3. Internet-delivered CBT (iCBT): Structured CBT programs primarily administered via online platforms or digital modules with remote therapist support. 4. Specialized Skills Training Groups: This category encompassed structured, manualized group interventions focusing on skill acquisition, primarily Dialectical Behavior Therapy Skills Training (DBT-ST) and other integrated CBT approaches targeting specific comorbidities (e.g., substance use disorder). This group was distinguished from general Group CBT by its intensive focus on specific skill modules.

This subgroup analysis was applied to the four primary outcome domains where sufficient data were available across multiple delivery modes: core ADHD symptoms, depressive mood, quality of life, and anxiety/stress.

3. Results

3.1. Literature screening process and outcomes

The search identified 16,195 records. Following deduplication and exclusion of non-RCTs, 3089 articles underwent title/abstract screening, with 162 proceeding to full-text review. After a secondary screening of these full texts, 69 articles were identified, and an additional 55 articles were excluded for not meeting the inclusion criteria. Ultimately, 14 articles were included for Meta-analysis. The detailed literature screening process is depicted in Fig. 1.

3.2. Quality assessment of included studies

Among the included studies, a total of 1223 adult ADHD patients were reported, with 611 in the experimental group and 612 in the control group. The basic characteristics of the included studies are presented in Tables 1 and 2.

All 14 included studies utilized randomization methods, with 3 studies (Van Emmerik-van Oortmerssen et al., 2019; Moritz et al., 2021; Nakashima et al., 2022) specifying the method of generating the random sequence, which is considered low risk; 11 studies (Nasri et al., 2023; Solanto et al., 2008; Huang et al., 2019; Cherkasova et al., 2020; Lam

Table 3
Included study outcome indicators.

Study	Nation	Outcome index
Van 2019	Netherlands	ARS, TLFB, BDI, BAI, EQ-5D
Nasri 2023	Sweden	ASRS, CGI-S, AAQoL, MADRS-, DERS, PSS, WAI-, SWLS, SD, EQ-5D index, EQ-VAS, EuroQol-5D, ISI
Solanto 2008	America	AISRS-IN, CAARS
Moritz 2021	Brazil	ADHD-RS, CAARS, BIS, SAS, SDS-1, STAI, STAIX-II, SES, GSES, WHOQOL-BREF, SDS-2
Huang 2019	China	ADHD-RS, CAARS, SAS, SDS, BRIEF-A, CANTAB, SES, WHOQOL-BRIEF
Cherkasova 2020	Canada	ASRS
Lam 2019	Germany	ADHD-RS (ADHD–Rating Scale)
Halmøy 2022	Norse	CSS, OAW, ISE, BDI, BAI, SCL-90
Pan 2022	China	CAARS, CGI, BDI
Nakashima 2022	Japan	BRIEF-A, DERS, AAQoL, BDI, BAI
Young 2015	Britain	ADHD-RS, QoL, EF
Virta 2010	Finland	CAARS-S-IN, CAARS-Ob-IN, CGI-S, SDS-W
Safren 2010	America	K-SADS, CGI, BCS, BAI, BDI
Corbisiero 2018	Switzerland	BADDS, ASRS, SCL-90, BDI-I, Q-LES-Q, CGI

Note: ARS-ADHD Rating Scale-ADHD; TLFB-Time Line Follow Back; BDI-Beck Depression Inventory; BAI-Beck Anxiety Inventory; EQ-5D-3-level EQ-5D-3; EQ-5D; ASRS-Adult ADHD Self-Report Scale-; CGI-S-Clinical Global Impression–Severity scale; AAQoL-Adult ADHD Quality of Life Questionnaire-; MADRS-S-Montgomery-Åsberg Depression Rating Scale (self-report version); DERS-Difficulties in Emotion Regulation Scale; PSS-4-Perceived Stress Scale–4 item version; WAI-1-Work Ability Index 1item; SWLS-Satisfaction With Life Scale; SDS-Sheehan Disability Scale; EQ-5D index-EuroQol-five dimensions Questionnaire-EuroQol; EQ-VAS-EuroQol-5D with Visual Analogue Scale-EuroQol-5D; ISI-Insomnia Severity Index; AISRS-IN-Adult ADHD Investigator Symptom Rating Scale - Inattentive symptoms; ADHD-RS-ADHD Rating Scale; CAARS-Conners Adult ADHD Rating Scales; BIS-Barratt Impulsiveness Scale; SAS-Self-rating Anxiety Scale; SDS-1-Self-rating Depression Scale; STAI-State-Trait Anxiety Inventory; STAIX-II-State-Trait Anger Expression Inventory-2; SES-Self-Esteem Scale; GSES-General Self-Efficacy Scale; WHOQOL-BREF-WHO Quality of Life-Brief Version; SDS-2-Sheehan Disability Scale; BRIEF-A-Behavior Rating Inventory of Executive Function-Adult Version; CANTAB-Cambridge Neuropsychological Testing Automated Battery; CSS-Current ADHD Symptoms Scale; OAW-Organization and Activation for Work; ISE-Index of Self-Esteem; SCL-90-Symptom Check List; CAARS-S-IN-Conners' Adult ADHD Rating Scale–Self-Report: Inattention/Memory subscale; CAARS-Ob-IN-Conners' Adult ADD Rating Scale–Observer Report: Inattention/Memory subscale; SDS-W-Sheehan Disability Scale WORK-Sheehan; K-SADS-Kiddie-Schedule for Affective Disorders and Schizophrenia; BADDS-Brown Attention Deficit Disorder Scale; Q-LES-Q-Quality of Life Enjoyment and Satisfaction Questionnaire; CGIS-Clinical Global Impression scale; SRAS-Self-report of ADHD symptoms; WRAADDS-Wender-Reimherr Adult Attention Deficit Disorder Scale-Wender-Reimherr.

et al., 2019; Halmøy et al., 2022; Pan et al., 2022) mentioned the use of random methods but did not provide detailed procedures, which is considered at uncertain risk; 3 studies (Van Emmerik-van Oortmerssen et al., 2019; Nasri et al., 2023; Young et al., 2015) mentioned the use of allocation concealment, and 10 studies (Van Emmerik-van Oortmerssen et al., 2019; Solanto et al., 2008; Huang et al., 2019; Lam et al., 2019; Halmøy et al., 2022; Pan et al., 2022; Nakashima et al., 2022; Virta et al., 2010; Safren et al., 2010; Corbisiero et al., 2018) mentioned the use of blinding, which are considered low risk; 7 studies were at low risk for detection bias (Van Emmerik-van Oortmerssen et al., 2019; Nasri et al., 2023; Solanto et al., 2008; Lam et al., 2019; Virta et al., 2010; Safren et al., 2010; Corbisiero et al., 2018); 1 study (Cherkasova et al., 2020) had a high risk of selective reporting bias. The specific risk details are shown in Fig. 2, Fig. 3.

3.3. Meta-analysis results of outcome measures

Among the 14 included studies, the outcome measures were assessed using seven categories of rating scales, which primarily included ADHD-related scales (ARS, ASRS, AISRS, ADHD-RS, CAARS, CSS, K-SADS, CGI-S), mood state scales (BDI, BAI, MADRS-S, SDS, BADDS), quality of life scales (AAQoL, SWLS, Q-LES-Q, WHOQOL-BREF, EQ-5D-3-level), stress and anxiety scales (DERS, PSS-4, STAI, STAIX-II), executive function scales (BRIEF, CANTAB, OAW), self-assessment scales (SAS, SES, ISE, GSES), and other scales (TLFB, EQ-5D index, EQ-VAS, ISI). Given the lack of uniformity in measurement methods or units across the included studies, the SMD was used as the pooled effect size. Therefore, this meta-analysis statistically evaluated the efficacy of different CBT interventions on various aspects of adult ADHD patients.

3.4. Heterogeneity assessment

Heterogeneity assessments for the primary outcome measures revealed considerable variation across the included studies (Table 2). Specifically, the analysis of core ADHD symptoms indicated moderate heterogeneity (I^2 = 41 %, P = 0.05). In contrast, the analyses for emotional state (I^2 = 0 %, P = 0.47), stress and anxiety (I^2 = 11 %, P = 0.34), and self-evaluation (I^2 = 27 %, P = 0.25) demonstrated low heterogeneity, suggesting a reasonable degree of consistency among the studies for these domains. Substantial heterogeneity was observed for quality of life (I^2 = 74 %, P = 0.0008) and executive function (I^2 = 65 %, P = 0.06), indicating significant variation in the results for these outcomes, which may be attributable to differences in intervention characteristics, sample demographics, or measurement tools.

3.5. Meta-analysis results

3.5.1. ADHD evaluation related scales

A total of 14 studies involving 559 participants in the CBT groups and 557 in the control groups were pooled for this outcome. Given the moderate heterogeneity (I^2 = 41 %), a random-effects model was employed. The meta-analysis demonstrated that CBT led to a significant reduction in core ADHD symptoms compared to control conditions, with a moderate effect size (SMD = −0.45, 95 % CI: −0.61 to −0.29; Z = 5.48, P < 0.00001) (Fig. 4).

3.5.2. The emotional state of depression scale

Nine studies comprising 413 participants in the experimental groups and 430 in the control groups were included. Due to low heterogeneity (I^2 = 0 %), a fixed-effects model was used. The analysis indicated a statistically significant, small-to-moderate improvement in depressive mood favoring CBT (SMD = −0.23, 95 % CI: −0.36 to −0.09; Z = 3.28, P = 0.001) (Fig. 5).

3.5.3. Quality of life scale

Seven studies involving 286 participants in the CBT groups and 298 in the control groups were analyzed. Substantial heterogeneity was present (I^2 = 74 %), leading to the use of a random-effects model. The pooled results showed no statistically significant difference in quality of life between the CBT and control groups (SMD = 0.12, 95 % CI: −0.21 to 0.46; Z = 0.73, P = 0.47) (Fig. 6).

3.5.4. Stress and anxiety scale

Seven studies involving 301 participants in the CBT groups and 316 in the control groups were analyzed. Given the low heterogeneity (I^2 = 11 %), a fixed-effects model was applied. The meta-analysis demonstrated a statistically significant, small-to-moderate reduction in stress and anxiety symptoms favoring CBT (SMD = −0.24, 95 % CI: −0.41 to −0.07; Z = 2.96, P = 0.003) (Fig. 7).

3.5.5. Executive function scale

Three studies comprising 154 participants in the experimental groups and 153 in the control groups were pooled. Although heterogeneity was low (I^2 = 27 %), a random-effects model was selected a priori

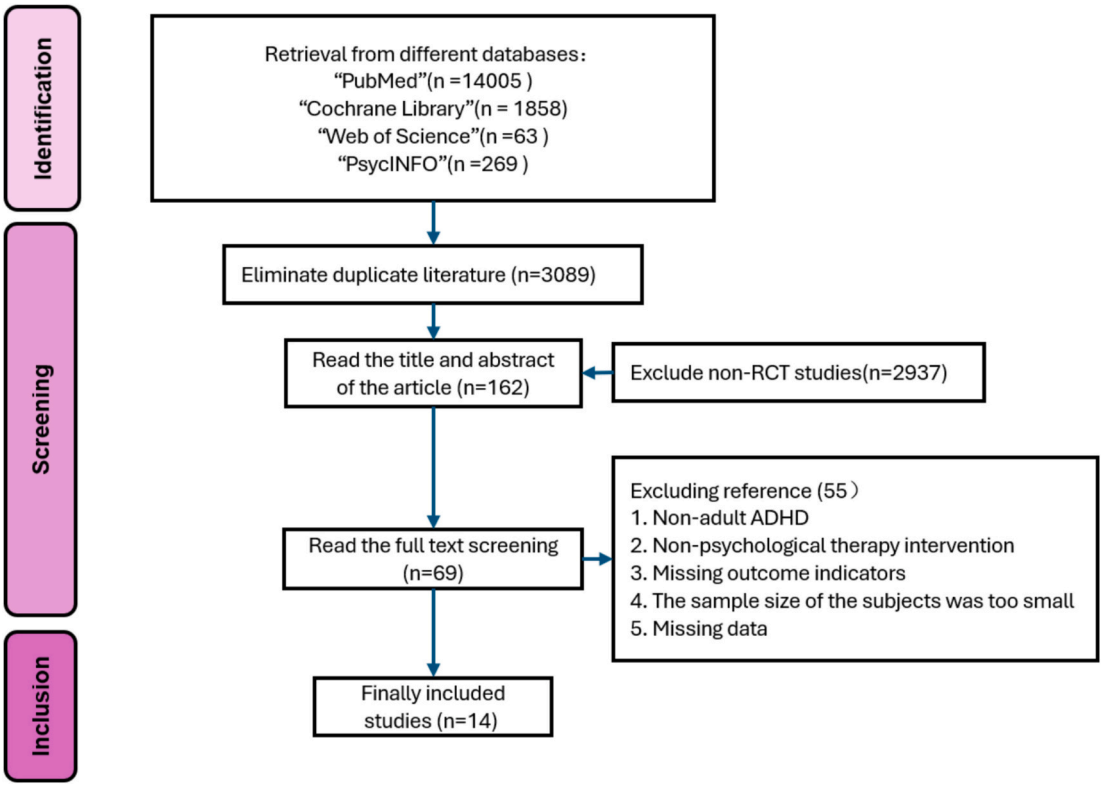


Fig. 1. Literature screening flow chart.

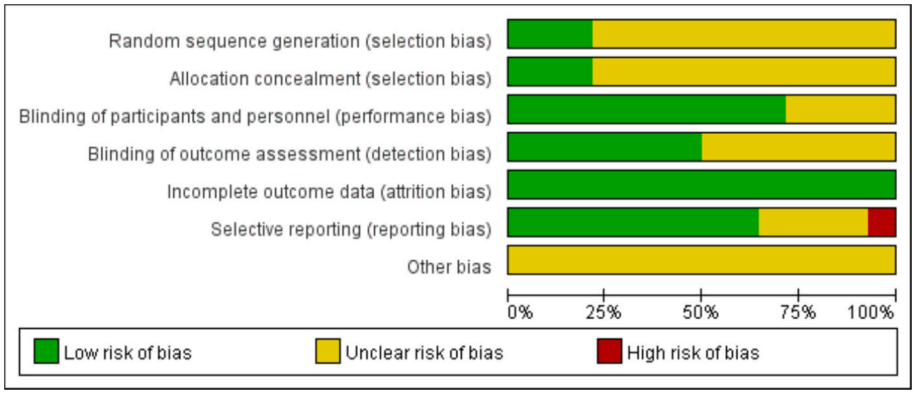


Fig. 2. Bias risk assessment chart of the quality of the 14 included studies.

due to the expected clinical and methodological diversity among the few included studies. The analysis revealed a significant, medium-sized improvement in executive function associated with CBT (SMD = −0.43, 95 % CI: −0.70 to −0.17; Z = 3.19, $P = 0.001$) (Fig. 8).

3.5.6. Self-evaluation scale

Four studies involving 179 participants in the CBT groups and 181 in the control groups were included. Substantial heterogeneity was observed ($I^2 = 81$ %), leading to the use of a random-effects model. The pooled results indicated no statistically significant difference in self-evaluation between the intervention and control groups (SMD = −0.30, 95 % CI: −0.79 to 0.18; Z = 1.24, $P = 0.22$) (Fig. 9).

3.6. Subgroup analysis

To explore potential sources of heterogeneity and compare intervention modalities, pre-specified subgroup analyses were conducted by categorizing studies into a 'pure CBT' group ($n = 9$ studies) and a 'CBT combination' group ($n = 5$ studies), the latter comprising CBT delivered alongside pharmacotherapy.

3.6.1. Subgroup analysis of ADHD outcome measures

Fourteen studies ($n = 563$ in CBT groups; $n = 553$ in control groups) were included in the analysis of core ADHD symptoms. Due to substantial heterogeneity ($I^2 = 70$ %, $P < 0.0001$), a random-effects model was applied. The overall analysis revealed a significant, small-to-moderate beneficial effect of CBT on core symptoms (SMD = −0.34, 95 % CI: −0.56 to −0.11; Z = 2.90, $P = 0.004$).

Subgroup analyses showed that this effect was driven by interventions using pure CBT (SMD = −0.52, 95 % CI: −0.71 to −0.33, $P < 0.00001$), as CBT combined with pharmacotherapy did not yield a significant effect (SMD = −0.08, 95 % CI: −0.52 to 0.37, $P = 0.74$) (Fig. 10).

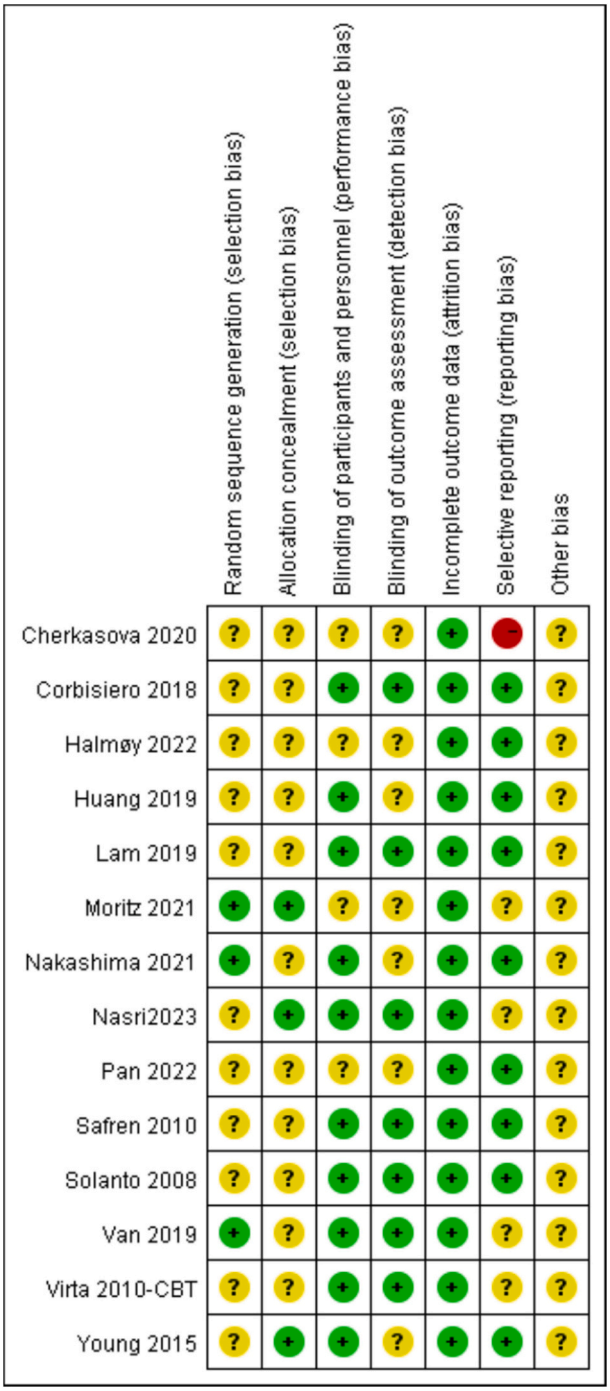


Fig. 3. Summary of the risk of bias in the quality of 14 articles included.

3.6.2. Subgroup analysis of depressive mood state indicators

For the outcome of depressive mood, nine studies involving 413 participants in the CBT groups and 430 in the control groups were analyzed. Heterogeneity among these studies was low ($I^2 = 0$ %, $P = 0.47$), supporting the use of a fixed-effects model. The pooled results indicated a statistically significant improvement in depressive symptoms favoring CBT over control conditions, with a standardized mean difference (SMD) of −0.23 (95 % CI: −0.36 to −0.09; Z = 3.28, $P = 0.001$).

Subgroup analyses showed that both the combined CBT and pharmacotherapy subgroup (SMD = −0.26, 95 % CI: −0.46 to −0.07; $P = 0.008$) and the pure CBT subgroup (SMD = −0.20, 95 % CI: −0.39 to −0.01; $P = 0.04$) contributed to this beneficial effect. According to Cohen's convention, the overall effect size (SMD = −0.23) represents a small-to-moderate improvement in depressive mood following CBT intervention in adults with ADHD (Fig. 11).

3.6.3. Subgroup analysis of quality of life indicators

Seven studies (total $n = 584$; CBT: 286, control: 298) were included in the analysis of quality of life (QoL). A random-effects model was employed due to substantial heterogeneity among the studies ($I^2 = 74$ %, $P = 0.0008$). The overall pooled analysis found no statistically significant improvement in QoL for CBT groups compared to control groups (SMD = 0.12, 95 % CI: −0.21 to 0.46; Z = 0.73, $P = 0.47$).

Subgroup analysis, however, revealed a distinct pattern: a significant improvement in QoL was observed in the subgroup receiving CBT combined with pharmacotherapy (SMD = 0.34, 95 % CI: 0.05 to 0.62; $P = 0.02$), whereas the subgroup receiving pure CBT showed no significant effect (SMD = 0.03, 95 % CI: −0.45 to 0.50; $P = 0.91$). According to Cohen's criteria, the overall non-significant effect size (SMD = 0.12) indicates a negligible effect of CBT on quality of life in adults with ADHD (Fig. 12).

3.6.4. Subgroup analysis of stress and anxiety indicators

Seven studies comprising 617 participants (CBT: $n = 301$; control: $n = 316$) were analyzed for outcomes on stress and anxiety. Heterogeneity was low ($I^2 = 11$ %, $P = 0.34$), and a random-effects model was applied. The meta-analysis showed a statistically significant reduction in stress and anxiety symptoms in the CBT group compared to the control group, with a standardized mean difference (SMD) of −0.24 (95 % CI: −0.41 to −0.07; Z = 2.77, $P = 0.006$).

Subgroup analysis revealed that the reduction was driven primarily by the subgroup receiving CBT in combination with pharmacotherapy (SMD = −0.53, 95 % CI: −0.83 to −0.24; $P = 0.0004$). In contrast, the subgroup receiving pure CBT alone did not show a statistically significant effect (SMD = −0.12, 95 % CI: −0.31 to 0.07; $P = 0.22$). According to Cohen's interpretation, the overall effect size (SMD = −0.24) indicates a small-to-moderate beneficial effect of CBT on stress and anxiety symptoms in adults with ADHD (Fig. 13).

3.6.5. Subgroup analysis of executive function indicators

The analysis of executive function included three studies, with 154 participants in the CBT groups and 153 in the control groups. Given the low heterogeneity ($I^2 = 27$ %, $P = 0.25$), a random-effects model was employed. The pooled results demonstrated a statistically significant improvement in executive function for the CBT groups compared to controls, with a standardized mean difference (SMD) of −0.43 (95 % CI: −0.70 to −0.17; Z = 3.19, $P = 0.001$).

Subgroup analysis indicated that this improvement was significant in the pure CBT subgroup (SMD = −0.56, 95 % CI: −0.84 to −0.28; $P < 0.0001$) but not in the CBT combined with pharmacotherapy subgroup (SMD = −0.18, 95 % CI: −0.58 to 0.22; $P = 0.37$). According to Cohen's criteria, the overall effect size (SMD = −0.43) represents a moderate beneficial effect of CBT on executive function in adults with ADHD (Fig. 14).

3.6.6. Subgroup analysis of self-evaluation indicators

Four studies (total $n = 360$; CBT: $n = 179$, control: $n = 181$) were included in the analysis of self-evaluation. A random-effects model was applied due to substantial heterogeneity among the studies ($I^2 = 81$ %, P = 0.001). The pooled analysis found no statistically significant difference in self-evaluation outcomes between the CBT and control groups (SMD = −0.30, 95 % CI: −0.79 to 0.18; Z = 1.24, $P = 0.22$).

Subgroup analyses showed no significant effects for either the combined CBT and pharmacotherapy subgroup (SMD = −0.14, 95 % CI: −1.05 to 0.76; $P = 0.76$) or the pure CBT subgroup (SMD = −0.47, 95 % CI: −1.04 to 0.10; $P = 0.11$). The overall effect size (SMD = −0.30) was small and not statistically significant, indicating that CBT did not produce a reliable improvement in self-evaluation in adults with ADHD

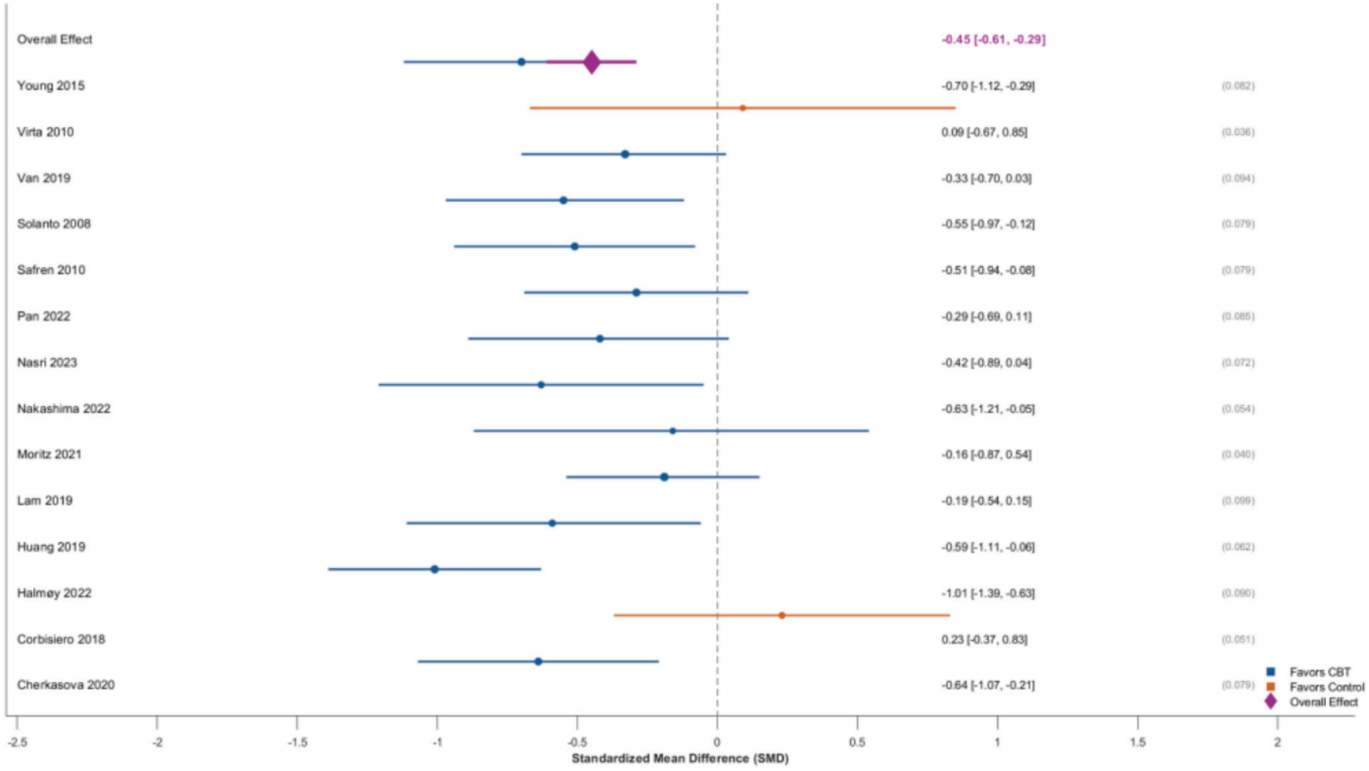


Fig. 4. Forest plot of the effect of CBT on the ADHD rating scale in adults with ADHD.

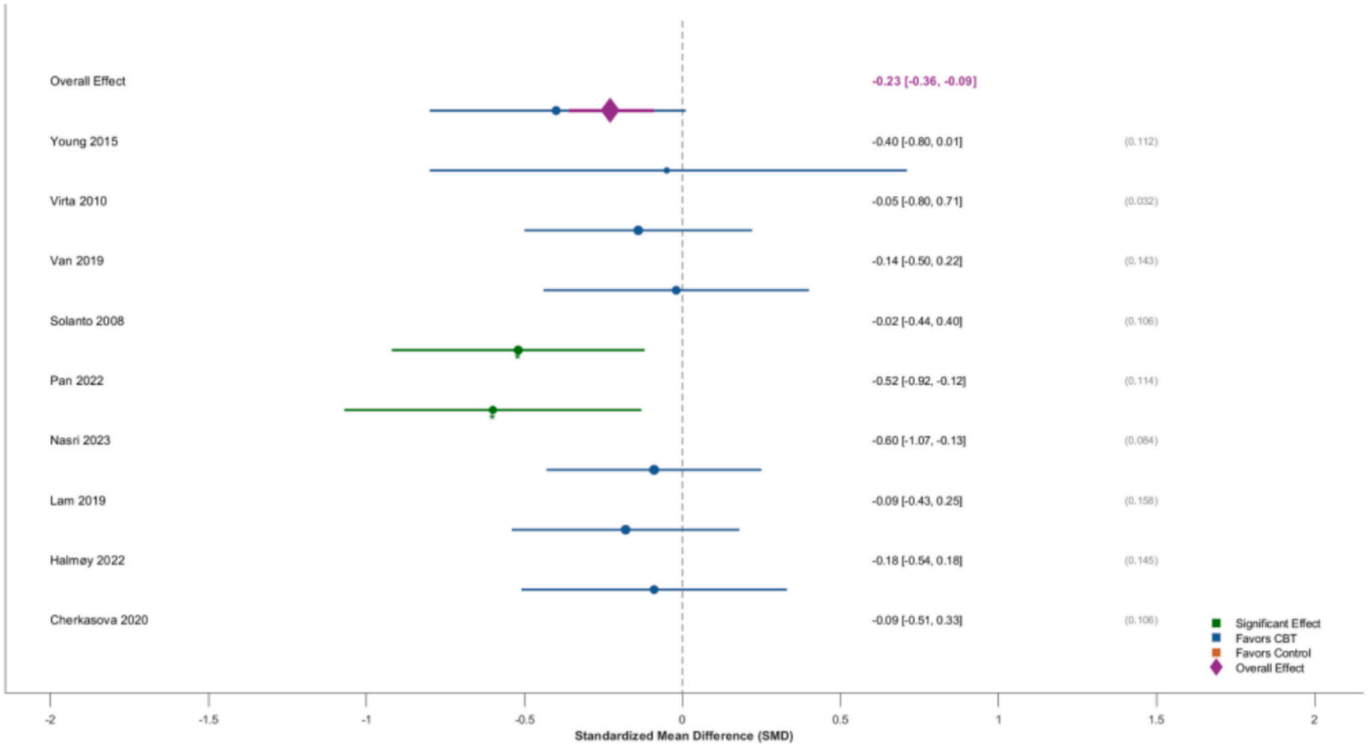


Fig. 5. Forest plot of the effect of CBT on the emotional state rating scale in adults with ADHD.

(Fig. 15).

3.7. Subgroup analysis by mode of CBT delivery

3.7.1. Effects on core ADHD symptoms by delivery mode

The subgroup analysis for core ADHD symptoms included 14 studies involving 563 participants in CBT groups and 553 in control groups. Due

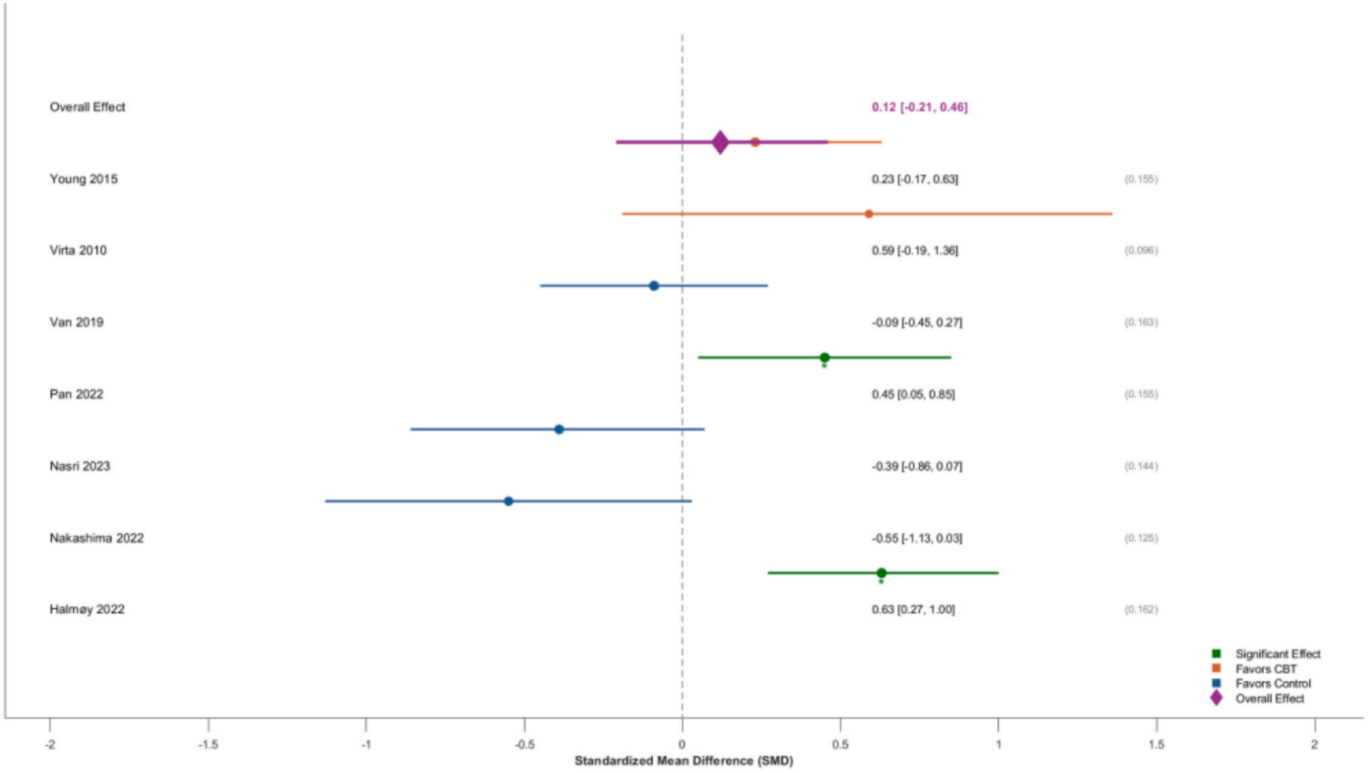


Fig. 6. Forest plot of the effect of CBT on the QOL rating scale in adults with ADHD.

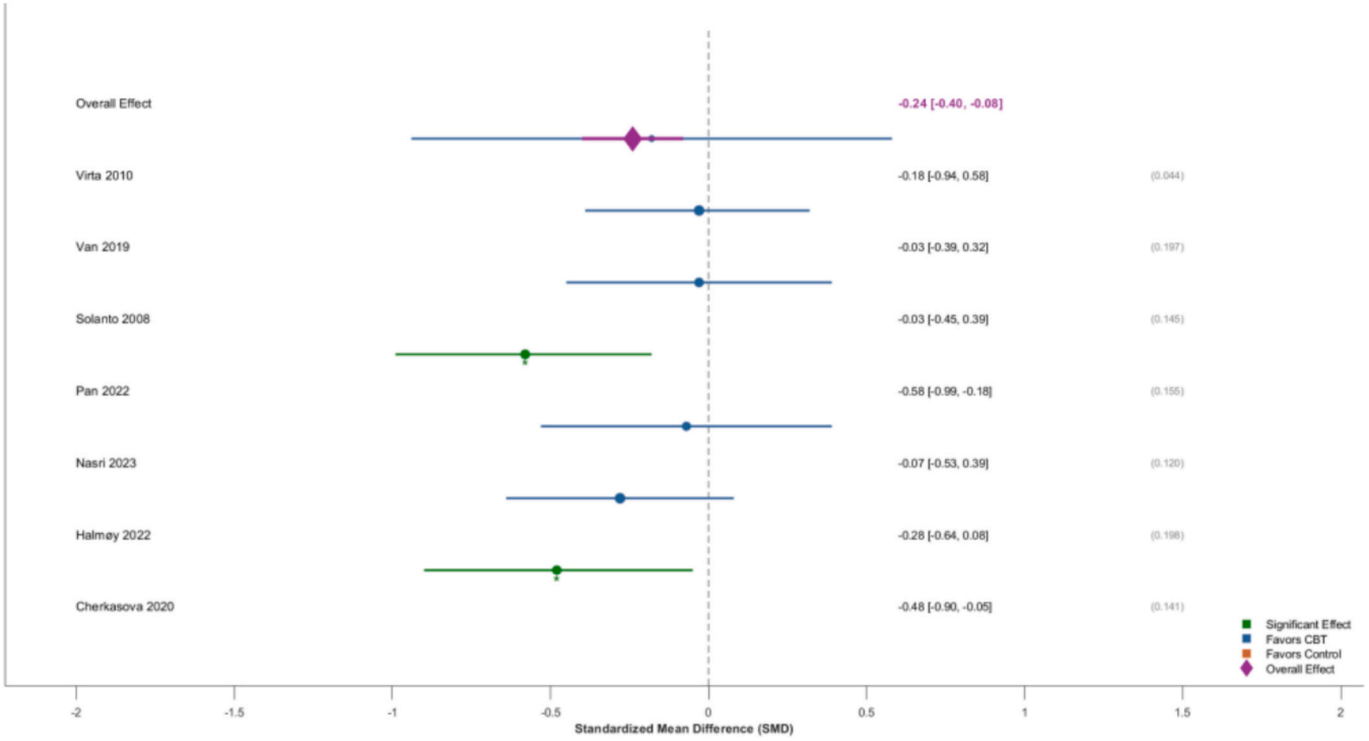


Fig. 7. Forest plot of the effect of CBT on the Stress and Anxiety Rating Scale in adults with ADHD.

to considerable heterogeneity among studies ($I^2 = 70$ %, $P < 0.0001$), a random-effects model was applied. The pooled analysis demonstrated a statistically significant overall improvement in core symptoms for CBT groups compared to controls (SMD = −0.34, 95 % CI: −0.56 to −0.11; Z = 2.90, $P = 0.004$) (Fig. 16).

While the test for subgroup differences was not statistically significant ($\chi^2 = 1.42$, df = 3, $P = 0.70$), a gradient of efficacy was observed. Group CBT showed the most significant improvement (SMD = −0.40, 95 % CI: −0.67 to −0.13; P = 0.004), followed by Specialized Skills Training Groups (SMD = −0.54, 95 % CI: −1.07 to −0.01; $P = 0.04$). The

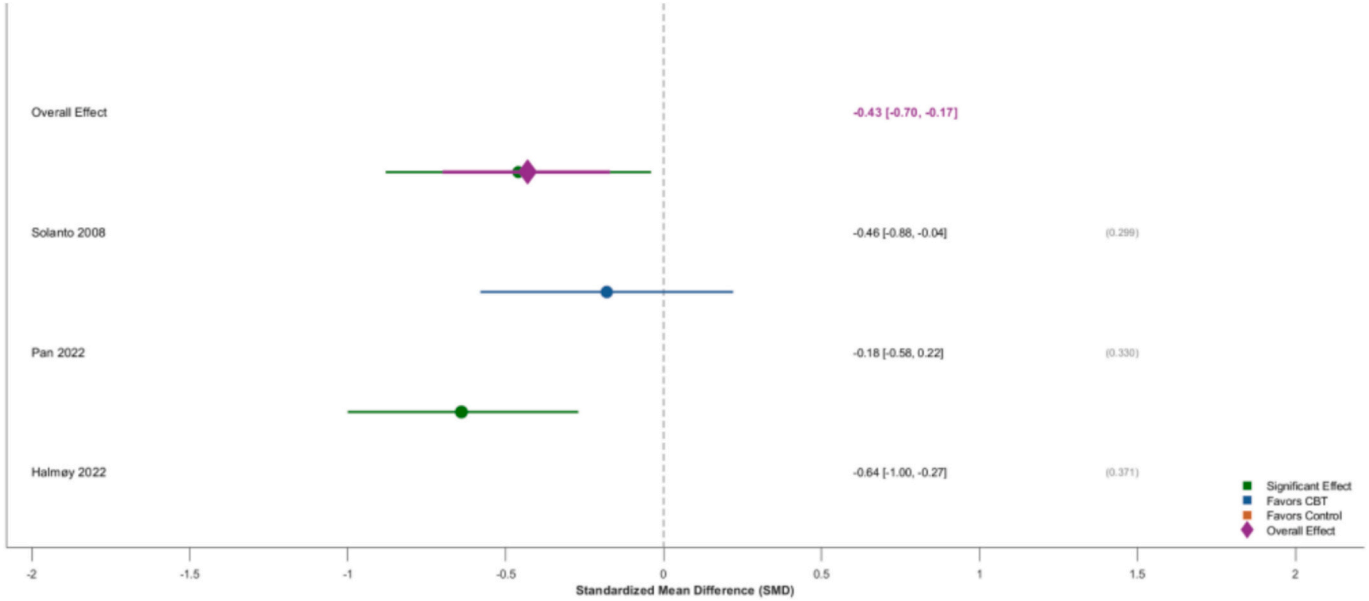


Fig. 8. Forest plot of the effect of CBT on the executive function rating scale in adults with ADHD.

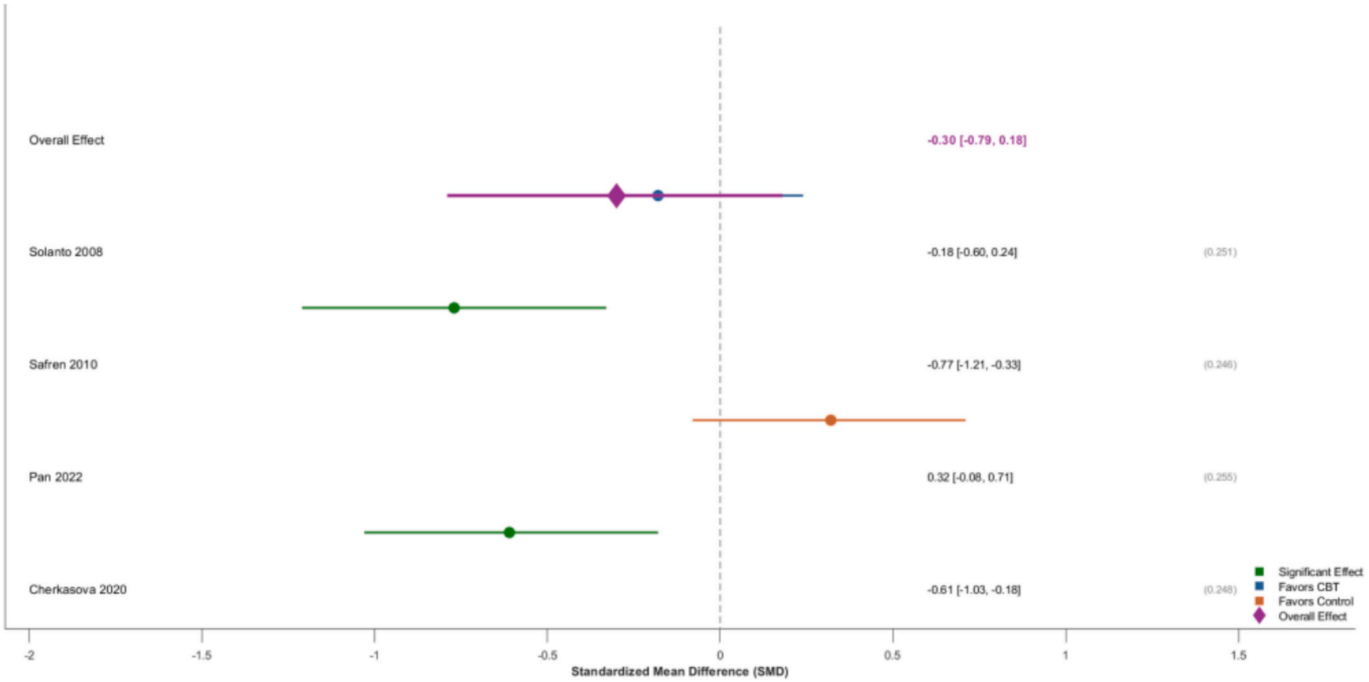


Fig. 9. Forest plot of the effect of CBT on self-rating measures in adults with ADHD.

pooled effect for Individual CBT was not statistically significant (SMD = −0.18, 95 % CI: −0.57 to 0.21; $P = 0.38$) and was characterized by high within-subgroup heterogeneity ($I^2 = 78$ %). Internet-delivered CBT, represented by a single study, indicated a non-significant trend toward symptom reduction (SMD = −0.42, $P = 0.08$). According to Cohen's criteria, the effects for both Group CBT and Specialized Skills Training represent a moderate degree of benefit, suggesting these modalities may offer more consistent positive effects for core ADHD symptoms in adults.

3.7.2. Effects on depressive mood by delivery mode

For depressive mood, nine studies comprising 413 CBT group participants and 430 control group participants were analyzed. Given low heterogeneity ($I^2 = 0$ %, $P = 0.47$), a fixed-effect model was used. Meta-analysis indicated a statistically significant small-to-moderate overall improvement in depressive symptoms favoring CBT (SMD = −0.23, 95 % CI: −0.36 to −0.09; Z = 3.28, $P = 0.001$) (Fig. 17).

Subgroup analysis revealed that Individual CBT (SMD = −0.32, 95 % CI: −0.54 to −0.09; $P = 0.006$) and Internet-delivered CBT (SMD = −0.60, 95 % CI: −1.07 to −0.13; $P = 0.01$) yielded significant positive effects. In contrast, Group CBT (SMD = −0.06, 95 % CI: −0.33 to 0.20; $P = 0.65$) and Specialized Skills Training Groups (SMD = −0.16, 95 % CI: −0.41 to 0.09; $P = 0.22$) showed no statistically significant effects. The test for subgroup differences was not significant ($\chi^2 = 4.79$, df = 3, $P = 0.19$), indicating no statistically demonstrable difference in efficacy across delivery modes for improving depressive mood.

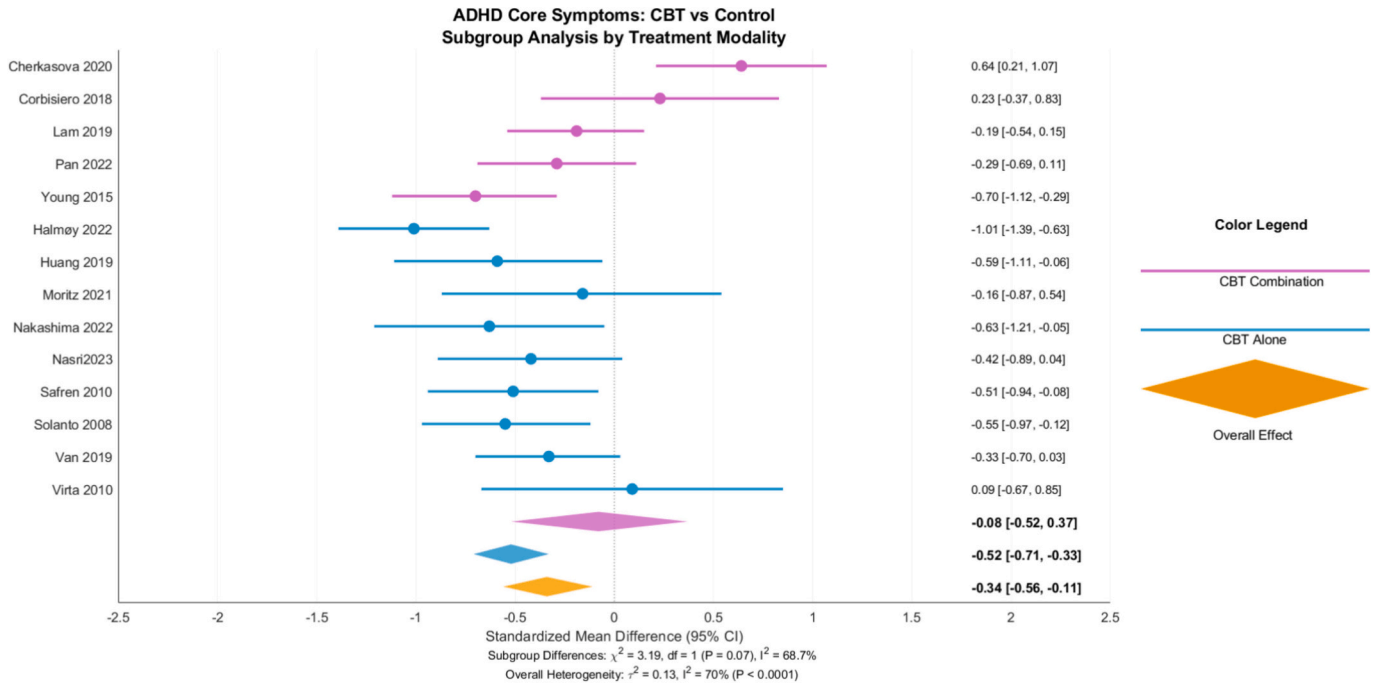


Fig. 10. Forest plots of the effects of CBT in different subgroups on the ADHD rating scale in adults with ADHD.

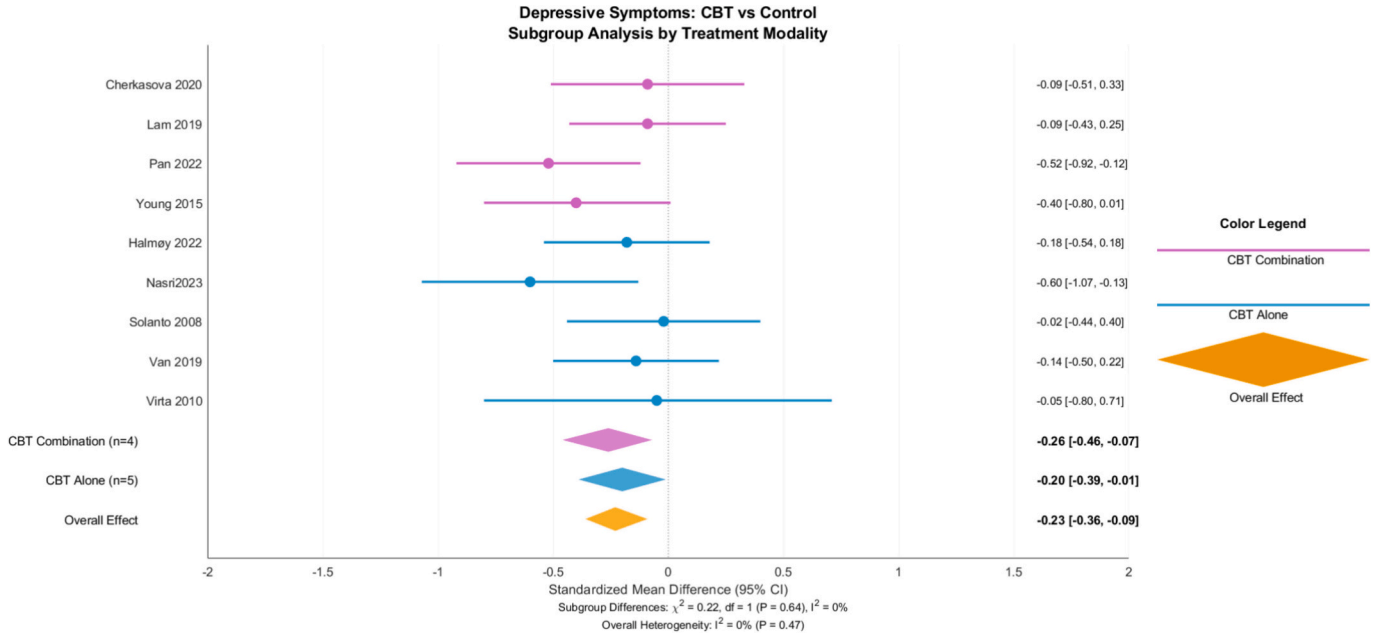


Fig. 11. Forest plots of the effects of CBT in different subgroups on the Depressive Emotional State rating Scale in adults with ADHD.

3.7.3. Effects on quality of life by delivery mode

Seven studies (CBT: $n = 286$, control: $n = 298$) were analyzed for quality of life (QoL). Substantial heterogeneity was present ($I^2 = 74$ %, $P = 0.0008$), warranting a random-effects model. The pooled analysis found no statistically significant overall effect of CBT on QoL (SMD = 0.12, 95 % CI: −0.21 to 0.46; Z = 0.73, P = 0.47) (Fig. 18).

However, subgroup analysis by delivery mode revealed a significant differential effect (Test for subgroup differences: $\chi^2 = 13.34$, df = 3, $P = 0.004$). Individual CBT demonstrated a significant, small-to-moderate positive effect on QoL (SMD = 0.37, 95 % CI: 0.10 to 0.64; $P = 0.007$). No significant effects were observed for Specialized Skills Training Groups (SMD = 0.27, 95 % CI: −0.44 to 0.98; $P = 0.45$), Internet-delivered CBT (SMD = −0.39, 95 % CI: −0.86 to 0.07; $P = 0.10$), or Group CBT (SMD = −0.55, 95 % CI: −1.13 to 0.03; $P = 0.06$). Notably, high heterogeneity was observed within the Specialized Skills Training subgroup ($I^2 = 87$ %). This result underscores that the specific delivery mode is a critical moderator of CBT's impact on QoL in adult ADHD.

3.7.4. Effects on anxiety and stress by delivery mode

The analysis for anxiety and stress included seven studies with 301 participants in CBT groups and 316 in control groups. Heterogeneity was low ($I^2 = 11$ %, $P = 0.34$), and a fixed-effect model was applied. A statistically significant small-to-moderate reduction in symptoms was

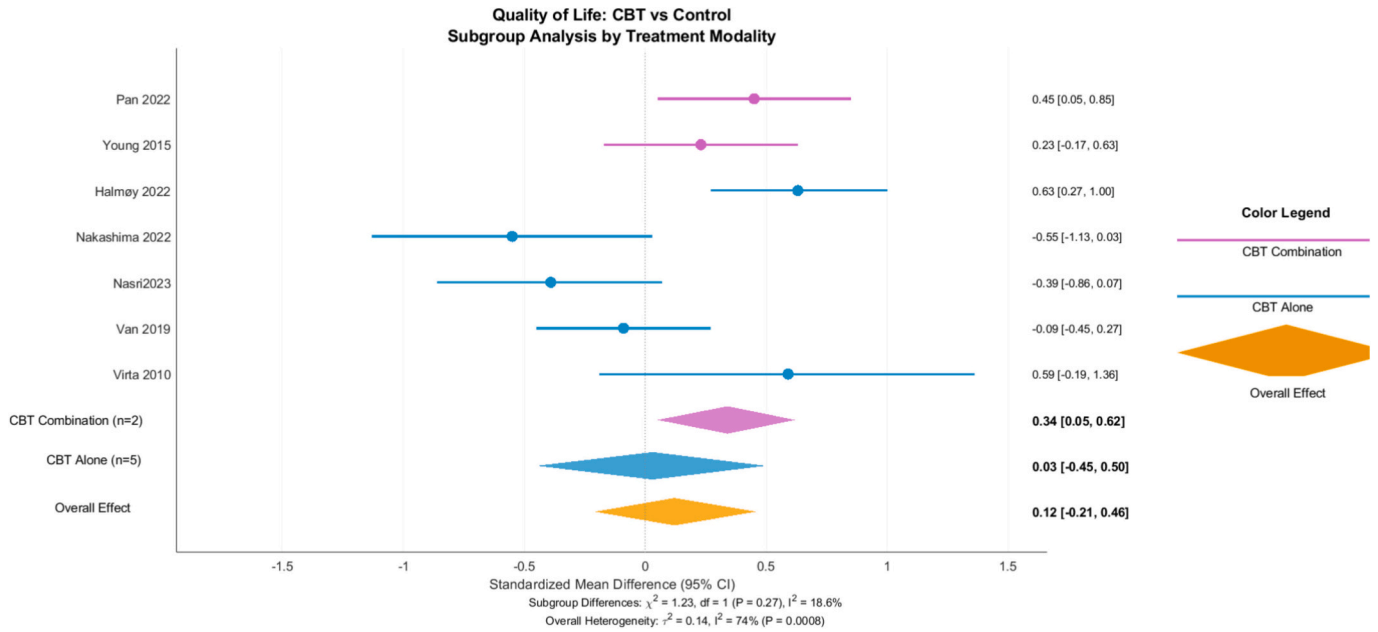


Fig. 12. Forest plots of the effects of CBT in different subgroups on the QOL rating scales for adults with ADHD.

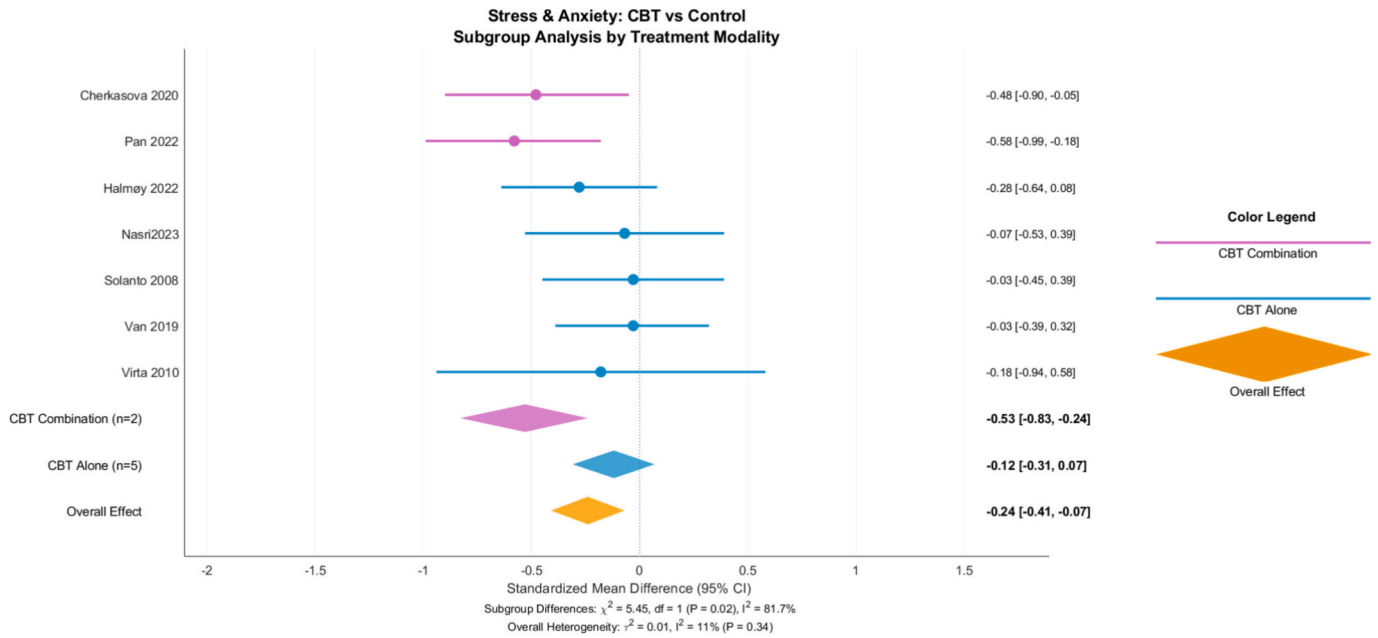


Fig. 13. Forest plots of the effects of CBT in different subgroups on the Stress and Anxiety Rating Scale in adults with ADHD.

found for CBT overall (SMD = −0.24, 95 % CI: −0.40 to −0.08; Z = 2.96, $P = 0.003$) (Fig. 19).

Subgroup analysis indicated a divergent pattern, though the test for subgroup differences was not statistically significant ($\chi^2 = 5.04$, df = 3, $P = 0.17$). Individual CBT showed a significant, moderate beneficial effect (SMD = −0.49, 95 % CI: −0.76 to −0.21; $P = 0.0005$). In contrast, no significant effects were observed for Group CBT (SMD = −0.03, 95 % CI: −0.45 to 0.39; $P = 0.88$), Internet-delivered CBT (SMD = −0.07, 95 % CI: −0.53 to 0.39; $P = 0.76$), or Specialized Skills Training Groups (SMD = −0.16, 95 % CI: −0.41 to 0.10; $P = 0.23$).

confirmed the robustness of the primary meta-analysis result for core ADHD symptoms (pooled SMD range: −0.47 to −0.41). Visual inspection of the funnel plot indicated slight asymmetry (Fig. 20), and Egger's regression test suggested potential publication bias (intercept = 1.82, $P = 0.03$). This asymmetry primarily involved a lack of small-sample studies with positive results, suggesting a possible overestimation of the effect size. However, the consistent direction and symmetric distribution of larger studies (SE < 0.3) support the reliability of the conclusion that CBT significantly improves core ADHD symptoms.

3.8. Bias of publication

Sensitivity analysis, performed by sequentially excluding each study,

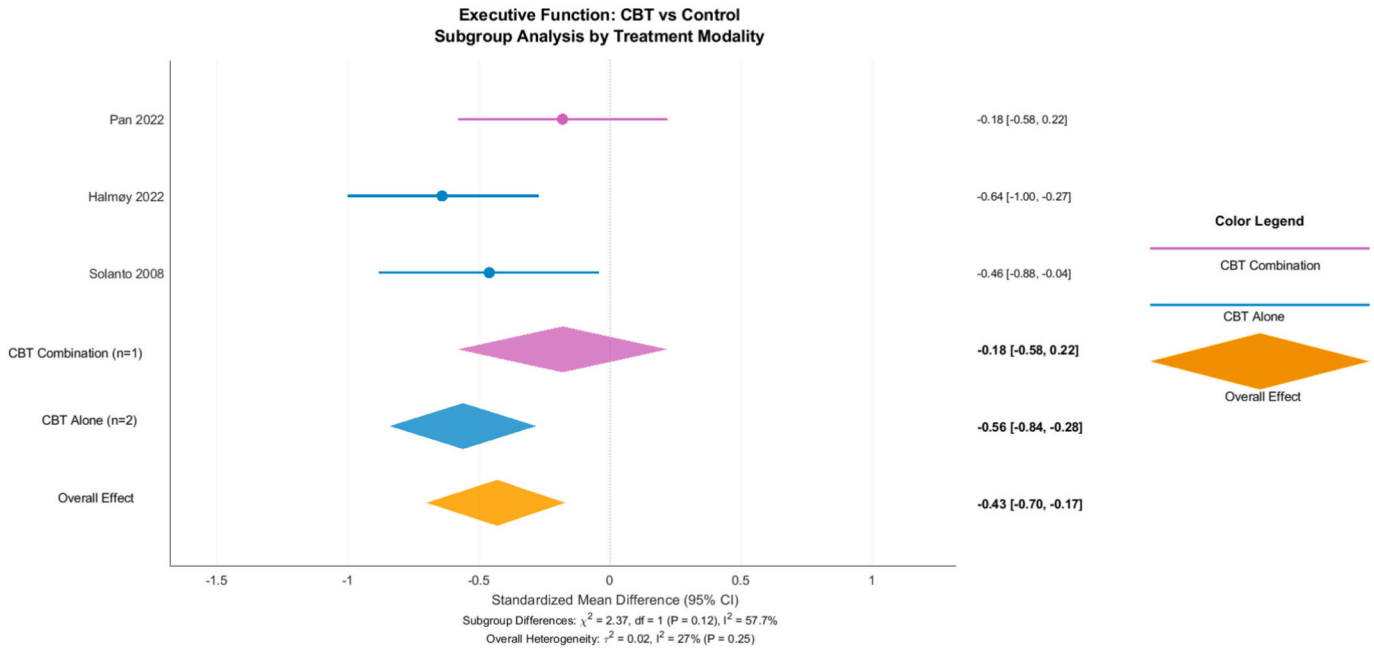


Fig. 14. Forest plot of the effect of CBT in different subgroups on the executive Functioning rating scale in adults with ADHD.

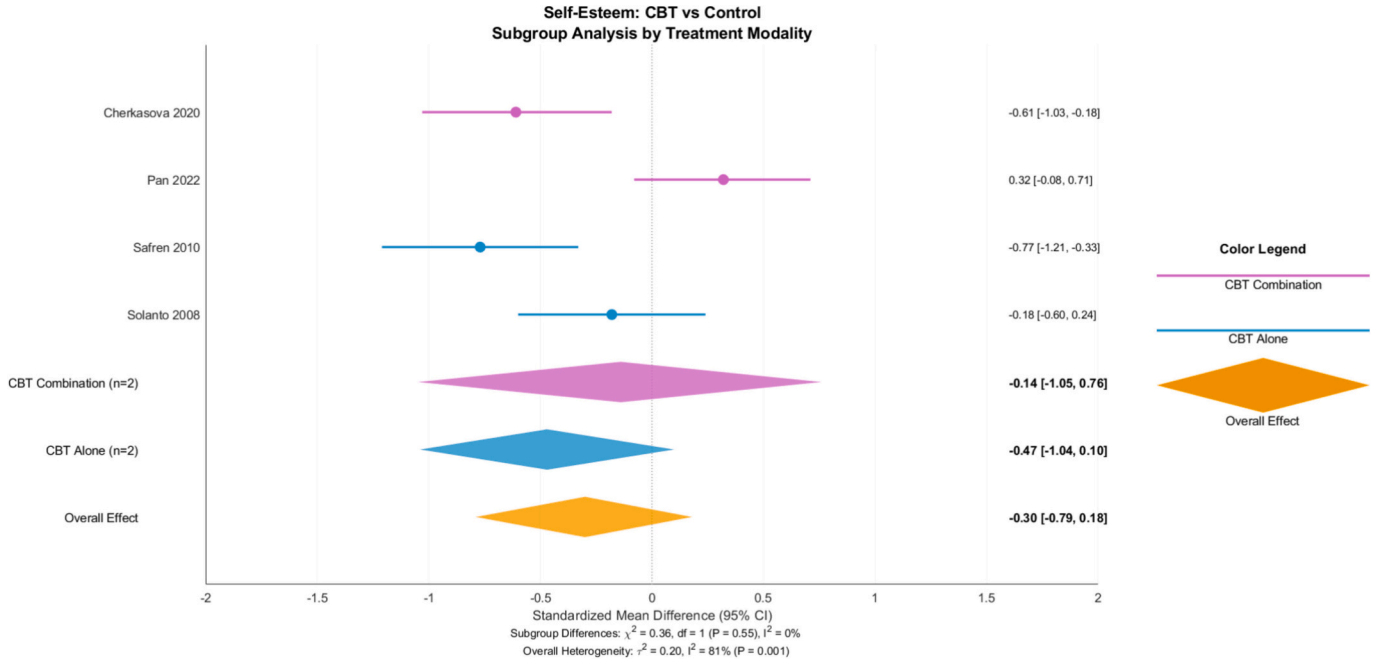


Fig. 15. Forest plots of the effect of CBT in different subgroups on self-rating measures in adults with ADHD.

4. Discussion

4.1. Therapeutic effects of CBT on adult ADHD

Our meta-analysis provides a nuanced evaluation of Cognitive Behavioral Therapy (CBT) for adult ADHD, demonstrating its differential efficacy across symptom domains. The findings confirm that CBT yields significant, small-to-moderate improvements in core ADHD symptoms (SMD = −0.45), depressive mood (SMD = −0.23), anxiety and stress (SMD = −0.24), and executive function (SMD = −0.43). These benefits are consistent with the proposed mechanism of CBT, which aims to modify maladaptive cognitive patterns and behaviors, thereby enhancing attention, self-control, and emotional regulation (Cherkasova et al., 2020; Van Emmerik-van Oortmerssen et al., 2019; Huang et al., 2019; Virta et al., 2010).

In contrast, CBT did not produce significant effects on quality of life (SMD = 0.12, $P = 0.47$) or self-evaluation (SMD = −0.30, $P = 0.22$). This dissociation suggests that ameliorating core symptoms and executive deficits may not automatically translate into perceived improvements in overall life satisfaction or self-concept. Achieving gains in these broader functional domains may require more comprehensive interventions that address environmental factors or incorporate targeted techniques for self-esteem, potentially over a longer duration (Young et al., 2015).

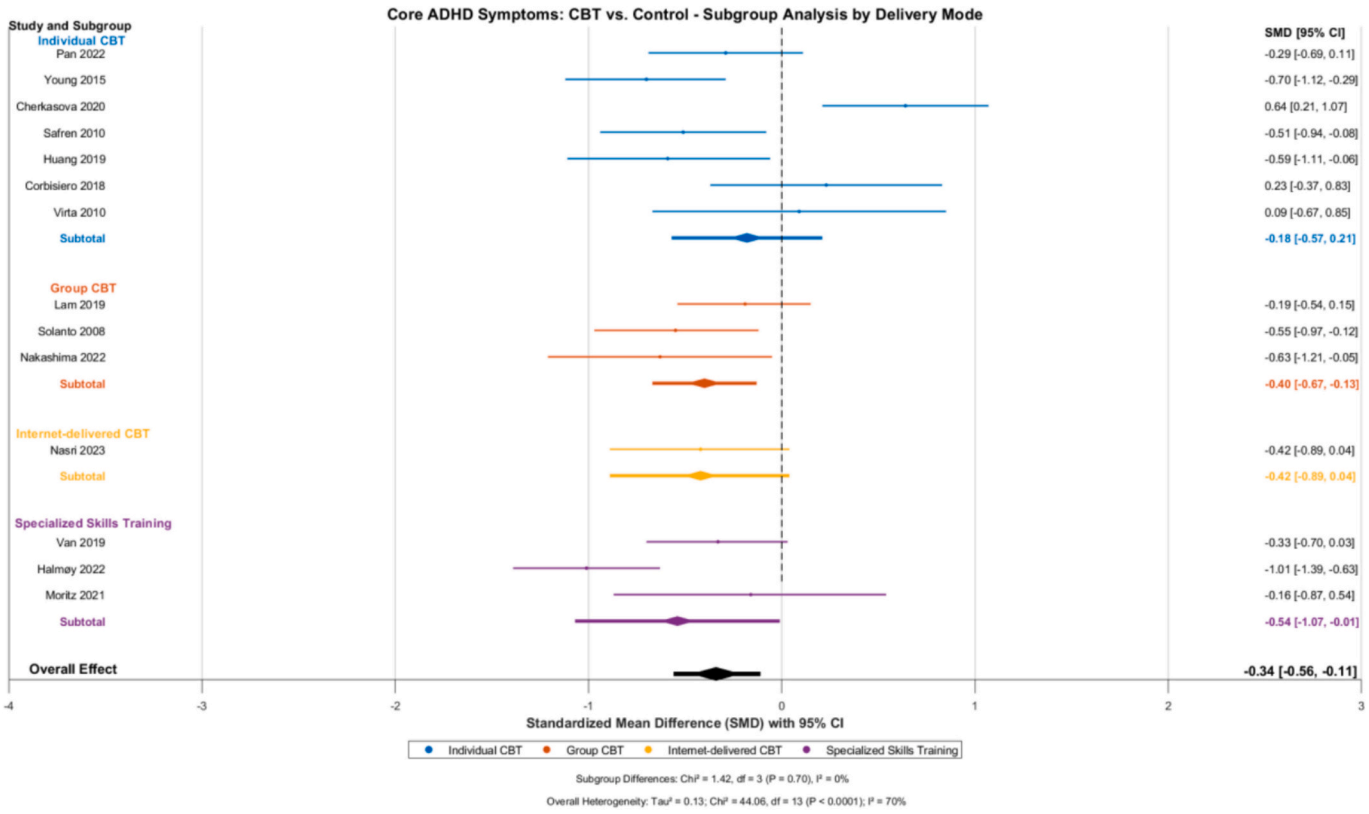


Fig. 16. Forest plot of subgroup analysis by delivery mode for the effect of CBT on core ADHD symptoms in adults.

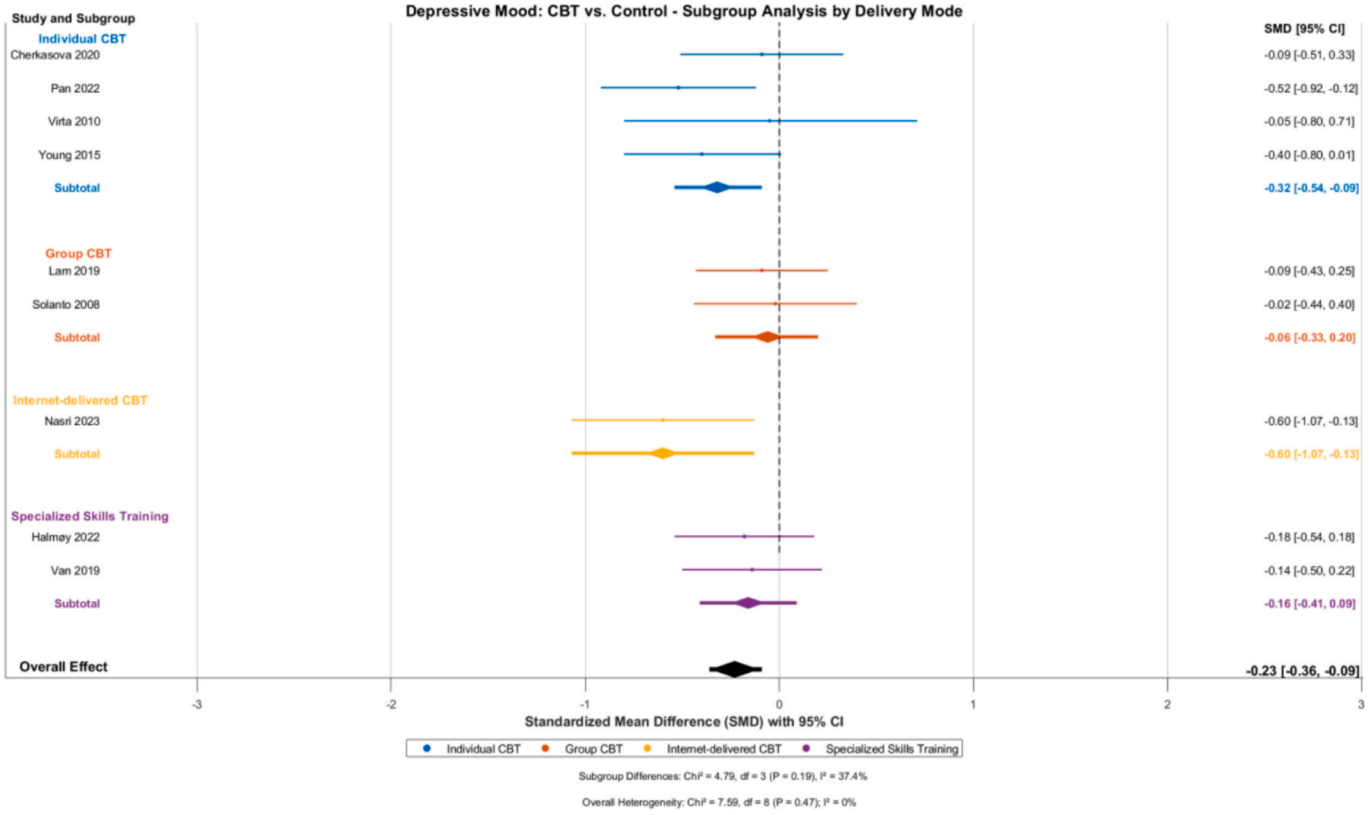


Fig. 17. Forest plot of subgroup analysis by delivery mode for the effect of CBT on depressive mood in adults with ADHD.

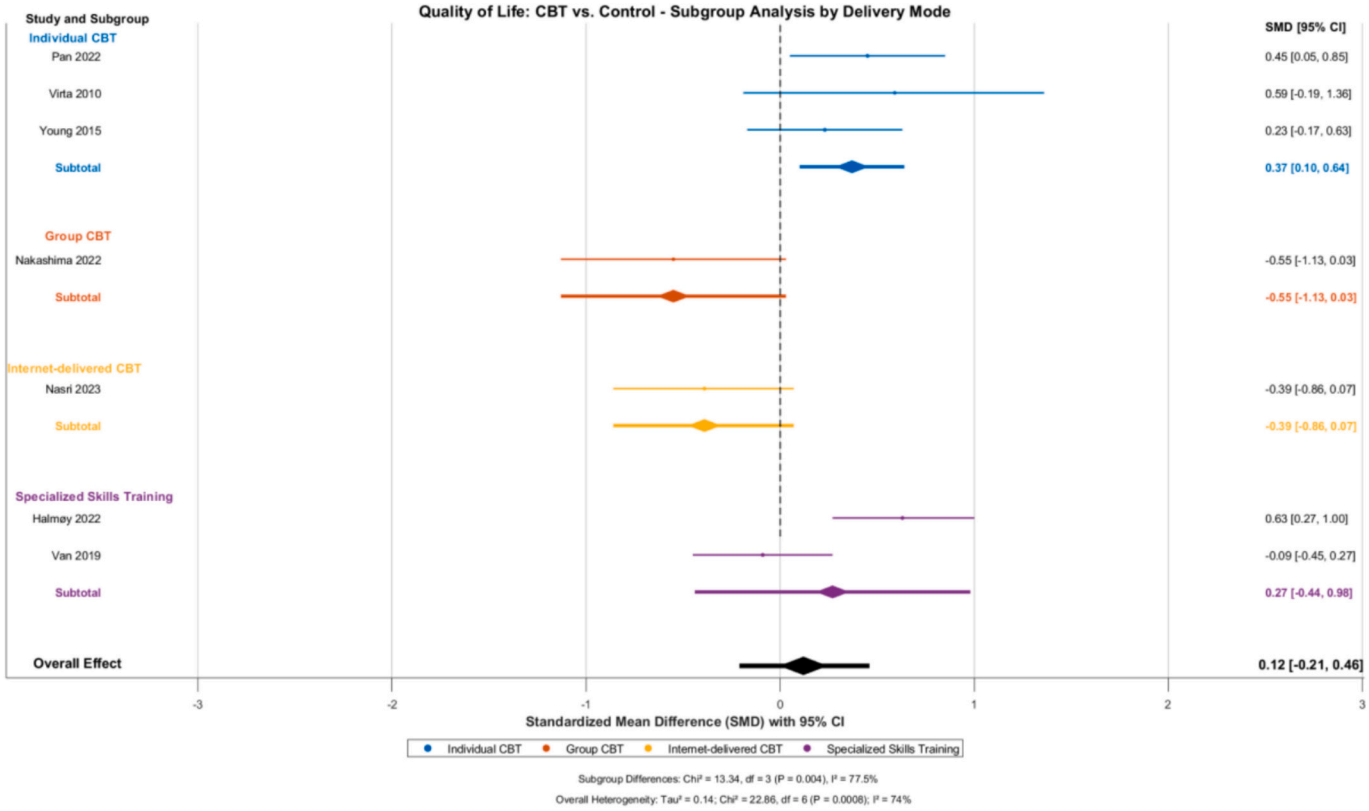


Fig. 18. Forest plot of subgroup analysis by delivery mode for the effect of CBT on quality of life in adults with ADHD.

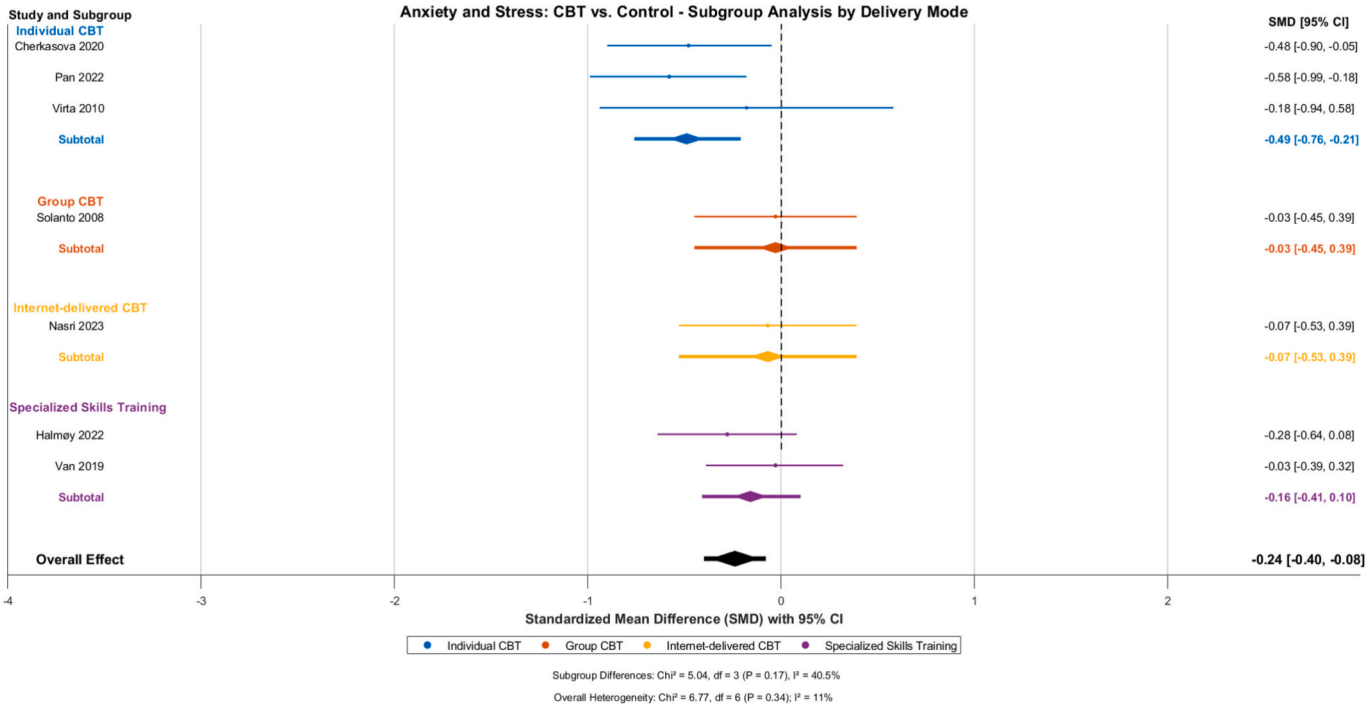


Fig. 19. Forest plot of subgroup analysis by delivery mode for the effect of CBT on anxiety and stress in adults with ADHD.

4.2. Combined CBT and pharmacotherapy

The integration of CBT with pharmacotherapy holds promise for a more robust and sustained intervention effect. This combined approach leverages complementary mechanisms: pharmacotherapy provides rapid neurochemical stabilization to alleviate core symptoms, while CBT equips patients with enduring cognitive, emotional, and behavioral self-management skills (Cherkasova et al., 2020; Lam et al., 2019; Pan et al., 2022). The synergy between symptom reduction through medication and the development of adaptive coping strategies through CBT may

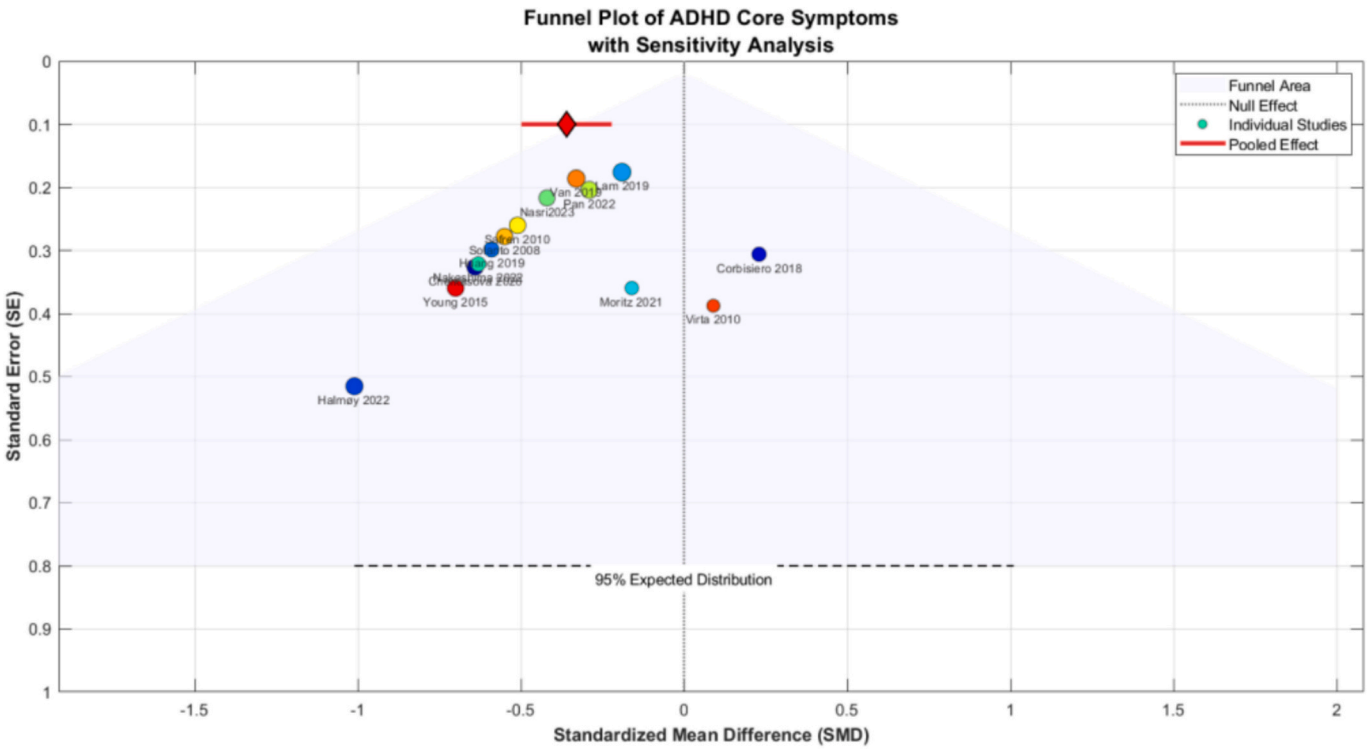


Fig. 20. Forest plots of the effect of CBT in different subgroups on self-rating measures in adults with ADHD.

enhance overall outcomes and reduce relapse risk.

4.3. Subgroup analysis of CBT effects

The interpretation of our findings must be contextualized within the substantial heterogeneity observed across included studies, which stemmed from three primary sources: intervention characteristics, participant demographics, and methodological approaches.

Intervention characteristics contributed significantly to variability in outcomes. The Dutch trial employing "CBT + substance use disorder treatment" (Van Emmerik-van Oortmerssen et al., 2019) demonstrated substantial core symptom improvement (SMD = −0.45), likely reflecting benefits from addressing comorbid conditions alongside ADHD. Conversely, the Swedish iCBT study (Nasri et al., 2023), while offering accessibility, may have provided less intensive executive function training due to the absence of real-time therapeutic feedback. Treatment duration also varied considerably, ranging from 8-week interventions (Nakashima et al., 2022) to year-long protocols (Lam et al., 2019). This variation aligns with neuroplasticity models suggesting that longer interventions are necessary to produce structural changes supporting quality-of-life improvements, potentially explaining the limited overall effect on this domain (SMD = 0.12).

Participant characteristics introduced additional heterogeneity. Studies with predominantly female samples (>60 %; Solanto et al., 2008; Nasri et al., 2023) typically emphasized emotional regulation outcomes, while those with male-dominated samples (Cherkasova et al., 2020, 70.1 % male) focused more on core symptoms. This likely explains the gender-based differential response to pure CBT for emotional symptoms (SMD = −0.20 for women vs. −0.12 for men). Age representation further complicated comparisons, with samples spanning 18–73 years (Young et al., 2015). Younger adults may show greater responsiveness to behavioral training due to enhanced neuroplasticity, while middle-aged participants (e.g., Halmøy et al., 2022, mean age 37) faced complex psychosocial challenges requiring broader support systems.

Methodological factors substantially influenced results. The utilization of 28 different assessment scales, most generic rather than ADHD-specific, presents a critical limitation. Only three studies used ADHD-specific quality-of-life measures (AAQoL; Nasri et al., 2023; Nakashima et al., 2022; Corbisiero et al., 2018), while others relied on generic instruments (e.g., EQ-5D) that may lack sensitivity to ADHD-specific impairments like time perception deficits or procrastination-related distress. This measurement inadequacy likely contributed to the pronounced heterogeneity in quality-of-life outcomes (I^2 = 74 %) and the observed "decoupling" between executive function improvement (SMD = −0.43) and quality-of-life gains.

Cultural considerations further complicated interpretation. East Asian studies (Huang et al., 2019; Pan et al., 2022) reported lower hyperactivity scores on the CAARS, possibly reflecting cultural norms that suppress overt expressions of hyperactivity. This suggests potential cross-cultural measurement bias in standard ADHD assessment tools.

The combination therapy paradox—superior efficacy for anxiety (SMD = −0.53) but not core symptoms—may reflect medication's dual role: creating neurochemical conditions that facilitate emotional regulation while potentially reducing motivation for strategy internalization. The superior outcomes from sequential intervention models (Pan et al., 2022's "12-week medication stabilization + 12-week CBT intensification") versus synchronous approaches (Cherkasova et al., 2020) suggest that timing critically influences treatment efficacy.

Publication bias assessment revealed funnel plot asymmetry (Egger's test intercept = 1.82, p = 0.03), indicating possible overestimation of effects. However, the stability of pure CBT effects (SMD = −0.52) across sensitivity analyses supports the robustness of this finding.

Several limitations warrant acknowledgment. The high heterogeneity, though explored, constrains generalizability. The focus on short-term outcomes limits understanding of maintenance effects. Future research should prioritize longer follow-up periods, validate ADHD-specific measures cross-culturally, and investigate optimal sequencing for combined treatments. Despite these limitations, CBT demonstrates compelling evidence as a first-line non-pharmacological intervention for adult ADHD, particularly for addressing emotional dysregulation, while combined approaches show specific advantages for anxiety and quality

of life when properly sequenced.

The pre-specified subgroup analyses based on the mode of CBT delivery further illuminate how intervention characteristics moderate treatment efficacy across distinct symptom domains of adult ADHD. Contrary to a uniform effect, the findings reveal a differential efficacy profile: interventions that were most effective for core neurocognitive symptoms were not necessarily optimal for emotional or functional outcomes.

For core ADHD symptoms, the significant benefits observed for Group CBT and Specialized Skills Training Groups, but not for Individual CBT, challenge the assumption that higher individualization invariably yields superior cognitive outcomes (Solanto et al., 2008). The structure, peer modeling, and repeated skill practice inherent in group formats may be particularly potent for instilling organizational strategies and improving behavioral regulation, core deficits in ADHD (Nakashima et al., 2022). The high heterogeneity within the Individual CBT subgroup ($I^2 = 78$ %) suggests that its effectiveness may be highly variable, depending on unmeasured factors such as therapist expertise or the specific modular focus (Safren et al., 2010; Huang et al., 2019).

In contrast, for emotional and functional outcomes—depressive mood, anxiety/stress, and quality of life—Individual CBT emerged as the most consistently effective modality (Pan et al., 2022; Cherkasova et al., 2020; Young et al., 2015). This pattern suggests that the personalized therapeutic alliance, tailored cognitive restructuring, and focused attention on emotional dysregulation in a one-on-one setting may be crucial for addressing the internalizing symptoms and life satisfaction deficits that often accompany ADHD (Liu et al., 2023). The singular efficacy of Individual CBT in improving quality of life (SMD = 0.37), despite a non-significant overall effect, underscores its unique value in translating symptom reduction into perceived life improvement (Pan et al., 2022; Young et al., 2015).

This pattern of differential efficacy—whereby group-based formats demonstrate superior effects on core symptoms, while individual therapy excels in promoting emotional well-being—carries direct clinical implications. It supports the adoption of a stepped-care or modular treatment strategy: group-based CBT or skills training can be effectively deployed as a first-line intervention for core ADHD symptoms, offering advantages in both cost-effectiveness and peer support (Lam et al., 2019). For patients with prominent comorbid anxiety, depression, or significant impairments in quality of life, augmenting with or transitioning to individual CBT should be actively considered (Corbisiero et al., 2018). The lack of significant effects observed for internet-delivered CBT across several outcomes, which may be partly attributable to the limited number of available studies, underscores the need for further research to optimize engagement and therapeutic support in digital formats (Nasri et al., 2023).

In summary, the delivery-mode subgroup analysis moves beyond establishing whether CBT works to provide preliminary evidence on how and for whom different formats work best. It emphasizes that "CBT" is not a monolithic intervention and that the optimal format depends critically on the primary treatment target. Future research should aim to prospectively test these modality-by-outcome interactions and examine patient-level moderators (e.g., baseline symptom severity, comorbidity profile) to guide truly personalized treatment selection.

4.4. Limitations and future directions

This study has several limitations. Firstly, although we attempted to conduct a subgroup analysis, due to the limitations of the original research report data, we were unable to systematically explore the potential impact of the intervention dose (such as the total duration of the sessions), the specific components of CBT (such as the proportion of executive function training and emotion regulation modules), and the individual characteristics of the participants (such as the severity of ADHD and specific comorbidities) on the therapeutic effect. This unobserved heterogeneity may be one of the reasons for the high and insignificant heterogeneity of some outcomes, such as quality of life and self-evaluation. Future research should focus on reporting intervention details and patient baseline data in a more standardized manner, so as to answer precision medicine questions about the optimal dose of CBT, core active ingredients, and target populations through more powerful methods such as individual participant data meta-analysis.

CRediT authorship contribution statement

Yang Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Feilong Zhu:** Writing – review & editing, Project administration, Formal analysis. **Yueyong Yu:** Software, Project administration. **Yiqi Song:** Conceptualization. **Hao Luo:** Project administration. **Baohua Xu:** Conceptualization. **Shengqi Lin:** Project administration. **Zhitong Ha:** Project administration. **Dongqing Kuang:** Project administration. **Yongtao Xie:** Project administration. **Zhexue Chen:** Project administration. **Yuanchun Ren:** Writing – review & editing, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Consent for publication

Not applicable.

Ethics approval and consent to participate

Not applicable.

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Declaration of competing interest

The authors declare no competing interests.

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Data availability

The data used to support the findings of this study are included within the article.

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